

Unit 2 — Bias and Fairness

Bias in data & models • Impact on decision making • Types of bias • Real-world cases •
Detecting & mitigating bias • Implementing bias detection and fairness

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Source: Bartneck et al., *An Introduction to Ethics in Robotics and AI* — §2.2 (ML & bias), §3.6.2 (bias testing), §4.3.4 (Justice: credit, courts, healthcare). The book treats bias briefly; the taxonomy, metrics, mitigation and code are standard material tagged +.

1. Understanding Bias in Data and Models



Figure 4.1 — Justitia: the symbol of justice
— impartial, weighing evidence

Bias & fairness

Bias = systematic, unfair skew in outcomes for certain people or groups. **Fairness** = absence of unjustified discrimination. The ethical principle at stake is **Justice**: AI should act justly and without bias.

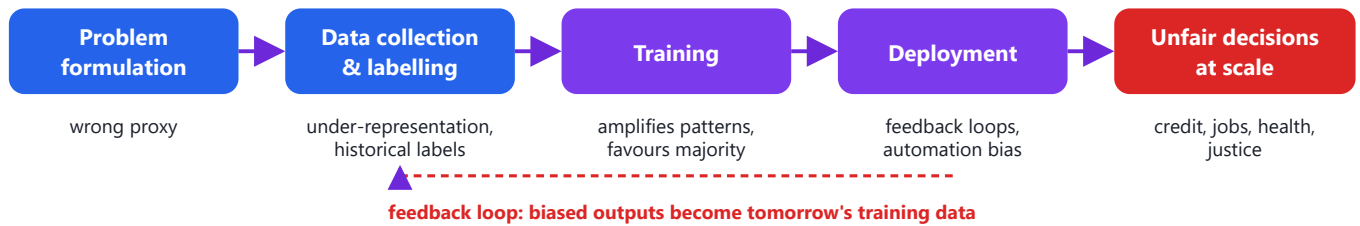
- **Entry point (book §2.2):** supervised learning needs **human-labelled data**, so human and societal biases can be baked in — a model trained on past decisions **reproduces the past**.

Where it comes from	
Data bias	Unrepresentative samples, skewed or prejudiced labels, missing groups, and proxy variables (neighbourhood ↔ race) that leak protected attributes +
Model bias	Poor problem formulation / objective (cost used as proxy for need), algorithm choices, optimising overall accuracy while a minority performs badly +

Removing the sensitive column is not enough

Other features act as **proxies** ("fairness through unawareness" fails). Blinding algorithms to race can even *hurt* (Kleinberg et al.), yet data must still reflect "what is the case". Trade-offs between fairness and accuracy must be made consciously.

2. How Bias Impacts the Decision-Making Process



- **Consequences:** unequal access to credit, jobs, healthcare and justice; **loss of trust**; violation of **justice** and **non-maleficence**; legal and reputational risk; one biased algorithm affects **millions**.
- **Automation bias:** people over-trust an "objective" score +. **Feedback loops:** predictive policing → more patrols → more arrests → "more crime data" +.

Book's discussion question

We expect AI to be fairer than humans — are there limits? Yes: it learns from human data, fairness has several **conflicting** definitions, and precise algorithms struggle with amorphous concepts.

3. Types of Bias +

Type	Meaning	Example
Historical	The world itself was unequal; data faithfully records it	Few women in past CEO data → model prefers men
Representation / sampling	Training sample doesn't reflect the population	Faces mostly light-skinned → poor accuracy on dark skin
Measurement / label	Proxy or labels measured inconsistently	Arrests as proxy for crime; cost as proxy for need
Selection	Non-random inclusion of data	Survey only of smartphone users
Algorithmic	Model or objective favours some groups	Optimising overall accuracy; ranking algorithms
Aggregation	One model for groups that differ	Single medical model for all ethnicities
Confirmation / cognitive	Developers' or users' beliefs shape design and reading of results	Choosing features that fit expectations
Automation	Over-reliance on algorithmic output	Judge defers to risk score
Deployment / feedback loop	Used differently than intended, or output changes future data	Predictive policing

4. Real-World Cases

Case	What happened	Lesson
COMPAS (recidivism, Broward County, USA)	ProPublica (2016): higher false-positive risk scores for Black defendants. Northpointe: scores are predictively fair given different base rates . Accuracy only ≈65% (Dressel & Farid). <i>Loomis v. Wisconsin</i> : defence couldn't inspect the proprietary algorithm; appeal failed.	Fairness definitions conflict (Chouldechova 2017: predictive parity ⊥ equal FPR/FNR when base rates differ); need for explainability

Healthcare risk algorithm (Obermeyer 2019)	Race excluded, but healthcare cost used as proxy for need; poor patients spend less (transport, childcare, work demands) → Black patients under-flagged; affects millions.	Problem formulation is the root cause; proxies encode structural inequality
Credit scoring	Applicant's neighbourhood data → systematic bias against certain residential areas; faster decisions scale unfairness.	Proxy variables
Amazon recruiting +	Trained on ~10 years of mostly male CVs; penalised CVs containing "women's"; scrapped (2018).	Historical bias
Face recognition +	<i>Gender Shades</i> (Buolamwini & Gebru 2018): far higher error for darker-skinned women.	Representation bias
Language models / chatbots +	Learn stereotypes and toxicity from web text (e.g. Microsoft Tay).	Training-data bias

5. Techniques to Detect Bias

1. **Audit the data** — group-wise counts, label rates, missing values; correlate features with protected attributes to spot **proxies**.
2. **Compare outcomes across groups** with fairness metrics (Section 6).
3. **Error analysis / slicing** — accuracy, FPR, FNR per subgroup and intersections (race × gender) +.
4. **Explainability** — feature importance, **SHAP / LIME** to expose proxy reliance +.
5. **Toolkits and audits (book §3.6.2)**: IBM **AI Fairness 360**, **audit-AI**, Facebook **Fairness Flow**, third-party audits (O'Neil Risk Consulting & Algorithmic Auditing), IEEE algorithmic-bias standards project (2017). Also Fairlearn and Google What-If Tool +.
6. **Diverse test cases / red-teaming** and **continuous monitoring** after deployment.

6. Fairness Metrics +

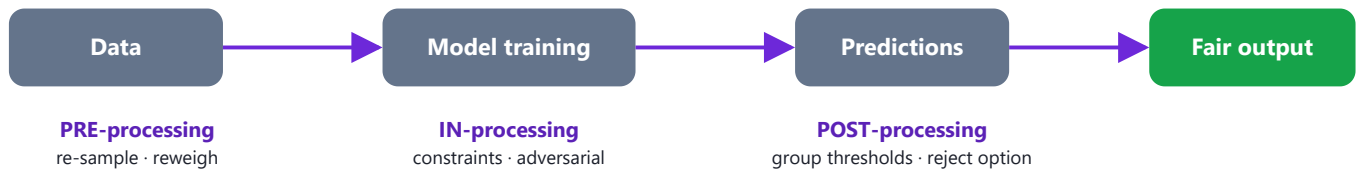
G = protected group, $\hat{Y}$ = prediction, Y = ground truth.

Metric	Requires	Idea
Statistical (demographic) parity	Equal selection rate	$P(\hat{Y}=1 \mid G=a) = P(\hat{Y}=1 \mid G=b)$; report the difference
Disparate impact ratio	Ratio ≥ 0.8 (80% rule)	selection rate of unprivileged ÷ privileged
Equal opportunity	Equal TPR	Qualified people approved at the same rate in every group
Equalised odds	Equal TPR and FPR	Both error rates equal across groups
Predictive parity	Equal precision	$P(Y=1 \mid \hat{Y}=1, G)$ same — COMPAS developers' choice
Calibration	Same risk per score	$P(Y=1 \mid \text{score}=s, G)$ same for all groups
Individual fairness	Similar → similar	Similar individuals receive similar outcomes
Counterfactual fairness	Invariance	Outcome unchanged if the protected attribute is flipped (causal)

Metrics cannot all hold at once

Unless base rates are equal or the classifier is perfect, you cannot satisfy them all (the COMPAS dispute). Choose the metric that fits the context and the harm.

7. Techniques to Mitigate Bias



Stage	Technique	Pros / cons
Pre-processing (fix the data)	More / representative data; re-sampling ; reweighing (weight each group×label cell so group $\perp$ label: $w = P(g) \cdot P(y) / P(g,y)$); remove or transform proxies; disparate-impact remover	Model-agnostic, cheap; may not remove proxy bias fully
In-processing (fix the learning)	Add fairness constraints / regulariser to the loss; adversarial debiasing (adversary tries to predict group from output); fair representation learning	Directly trades fairness vs accuracy; needs custom training
Post-processing (fix the output)	Group-specific thresholds ; equalised-odds post-processing; reject-option classification (relabel near-boundary cases)	Works on black-box models; can be legally sensitive

Non-technical measures

Diverse development teams • fairness requirements agreed with stakeholders • **datasheets for datasets** and **model cards** • **human-in-the-loop** review • appeal and redress mechanisms • regular **audits** • legal compliance (GDPR, anti-discrimination law) • honest handling of the **fairness** ↔ **accuracy** trade-off.

8. Implementing Bias Detection and Fairness

Runnable code: `UNIT-2/bias_detection_demo.py` (numpy + pandas + scikit-learn). Synthetic loan data: **group B was historically penalised**; `zip_score` is a **proxy** for group; the protected attribute itself is *not* a model feature.

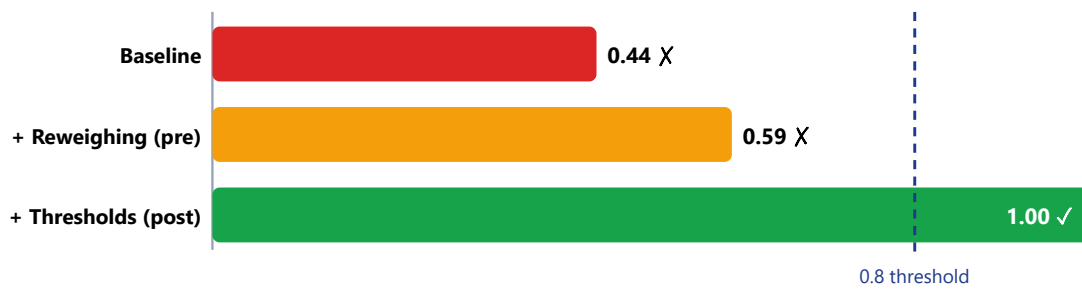
Train baseline → Measure per-group metrics → Mitigate (pre / post) → Re-measure

```
# 1. Detect: per-group selection rate, TPR, FPR (y_true = truly qualified)
sel_rate_B = y_pred[g == "B"].mean(); sel_rate_A = y_pred[g == "A"].mean()
disparate_impact = sel_rate_B / sel_rate_A # < 0.8 fails the 80% rule
tpr = y_pred[(g == grp) & (y_true == 1)].mean() # equal-opportunity check

# 2. Mitigate (pre-processing): reweighing w(g, y) = P(g) * P(y) / P(g, y)
model.fit(X_tr, y_tr, sample_weight=w)

# 3. Mitigate (post-processing): group-specific threshold equalises selection rate
thr_B = np.quantile(p_B, 1 - selection_rate_A)
```

Demo result — disparate impact ratio ($B \div A$); 0.8 = legal 80% rule



Model	Selection rate A / B	Disparate impact (B/A)	TPR gap (B-A)	FPR of B
Baseline	0.364 / 0.159	0.44 X	-0.45	0.000
+ Reweighing (pre)	0.331 / 0.197	0.59 X	-0.30	0.000
+ Thresholds (post)	0.364 / 0.364	1.00 ✓	+0.03	0.044

Takeaways

Dropping the protected column does **not** remove bias (the proxy leaks it); reweighing helps only partly; post-processing equalises outcomes but at a cost (FPR for B rose $0 \rightarrow 0.044$). Always report **several metrics plus accuracy**.

Library route (production): `pip install fairlearn aif360`.

```
from aif360.algorithms.preprocessing import Reweighing # IBM AI Fairness 360
from aif360.metrics import BinaryLabelDatasetMetric
from fairlearn.metrics import MetricFrame, demographic_parity_difference
from fairlearn.reductions import ExponentiatedGradient, DemographicParity
```

9. Quick Revision & Likely Questions

One-liners

- **Bias** = systematic unfairness; enters via **data** (historical, sampling, labels, proxies) and **model / objective**. Removing the protected attribute $\neq$ fair.
- **Impact**: unequal decisions in credit, hiring, health, justice at scale $\rightarrow$ loss of trust, legal and reputational risk.
- **Cases**: COMPAS (conflicting definitions, 65% accuracy, *Loomis*) • healthcare cost proxy (Obermeyer) • credit / neighbourhood • Amazon hiring • Gender Shades.
- **Detect**: data audit • parity / disparate impact (80%) / equal opportunity / equalised odds / predictive parity • subgroup errors • AIF360, Fairlearn, audits.
- **Mitigate**: **pre** (re-sample, reweigh) • **in** (constraints, adversarial) • **post** (thresholds, reject option) + diverse teams, documentation, oversight.
- Fairness metrics **conflict** $\rightarrow$ context decides; the fairness-accuracy trade-off is a stakeholder decision.

Likely questions

1. Define bias; how can it arise in data and in models? (5)
2. Explain how bias affects decision making using two real-world cases. (8)
3. List and explain the types of bias with examples. (8)
4. Describe the COMPAS controversy; why can't predictive parity and equal error rates coexist? (8)
5. Compare pre-, in- and post-processing mitigation techniques. (10)
6. Define statistical parity, disparate impact and equal opportunity; compute them from a confusion matrix. (6)
7. Describe/implement a procedure to detect and mitigate bias in a classifier. (10)